Supplementary Mathematical Appendix

The Volatility Penalty in Progressive Electricity Tariffs:
Sharp Finite-Sample Bounds and Constrained Tariff Design

Supplement to the IJSSER manuscript

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Numbering. Propositions 1–4, Theorems 1–3 and Corollary 1 carry the same numbers as in the manuscript; results numbered S$k.j$ appear only in this supplement.

Status of results. Results labelled *known* are classical and are credited to their sources. Results labelled *proved here* have complete proofs below; to the best of a limited literature search they are not stated in this form elsewhere, but no priority claim is made. Items labelled *conjecture* are supported numerically only. Population moments (divisor n) are used throughout; Remark 3 discusses the sample standard deviation.

Contents

S1 Definitions, notation and assumptions **1**

S2 Non-negativity, decomposition and unequal periods **2**

S3 Exact absolute-deviation and minority-side representations **3**

S4 Moment bounds **3**

S5 Finite-sample sharp bound (Theorem 1) **4**

S6 Threshold sensitivity (Proposition 4) **4**

S7 Non-negative consumption (Theorem 2) **5**

S8 Multi-tier tariffs: decomposition and the joint bound (Theorem 3) **5**

S9 Constrained tariff design **6**

S10 Extremal configurations and edge cases **6**

S11 Counterexamples to invalid generalisations **7**

S12 Numerical validation **7**

S13 Additional derivations and methodological details **7**

S1 Definitions, notation and assumptions

Definition 1 (Marginal-block tariff)**.** For thresholds $0 < K_1 < \cdots < K_m$ and marginal prices $p_0, \ldots, p_m \in \mathbb{R}$, with increments $\Delta p_j = p_j - p_{j-1}$, the energy charge for consumption $x \geq 0$ in

a reference period of 30 days is

$$B(x) = p_0 x + \sum_{j=1}^{m} \Delta p_j \,(x - K_j)_+ , \qquad (z)_+ = \max(z, 0). \tag{S1.1}$$

The tariff is *progressive* (convex) if $\Delta p_j \geq 0$ for all j. Writing $c_0 = 0$ and $c_k = \sum_{j \leq k} \Delta p_j K_j$, on block k (i.e. $K_k \leq x \leq K_{k+1}$, with $K_0 = 0,\ K_{m+1} = \infty$) we have $B(x) = p_k x - c_k$, and if the tariff is convex then

$$B(x) = \max_{0 \leq k \leq m} (p_k x - c_k) \qquad \text{for all } x \geq 0. \tag{S1.2}$$

Proof of (S1.2). The function $\ell_k(x) = p_k x - c_k$ satisfies $\ell_k(x) - \ell_{k-1}(x) = \Delta p_k (x - K_k)$, which is ≥ 0 iff $x \geq K_k$ when $\Delta p_k > 0$. Hence for x in block k, $\ell_k(x) \geq \ell_{k-1}(x) \geq \cdots \geq \ell_0(x)$ and $\ell_k(x) \geq \ell_{k+1}(x) \geq \cdots \geq \ell_m(x)$. If some $\Delta p_k = 0$ the corresponding pieces coincide. □

Definition 2 (Volatility penalty)**.** For $x = (x_1, \ldots, x_n)$ in n comparable periods, $\bar{x} = \frac{1}{n}\sum_t x_t$,

$$V(x) = \sum_{t=1}^{n} B(x_t) - nB(\bar{x}), \qquad W_K(x) = \sum_{t=1}^{n} (x_t - K)_+ - n\,(\bar{x} - K)_+ .$$

Write $s^2 = \frac{1}{n}\sum_t (x_t - \bar{x})^2$, $\mathrm{MAD} = \frac{1}{n}\sum_t |x_t - \bar{x}|$, $d = \bar{x} - K$ and $d_j = \bar{x} - K_j$.

Assumptions. (A1) Telescopic (marginal-block) billing. (A2) Fixed charges independent of consumption; they cancel from V. (A3) Consumption is held fixed (no behavioural response). (A4) Thresholds refer to a 30-day period; for other lengths they are prorated (Section S2). (A5) The comparator $nB(\bar{x})$ is hypothetical.

Remark 1 (All-units tariffs). An all-units (non-telescopic) schedule $B^{\mathrm{AU}}(x) = p_{k(x)} x$ is discontinuous and not convex; none of the results below apply (Section S10).

S2 Non-negativity, decomposition and unequal periods

Proposition 1 (Decomposition and non-negativity; Jensen [1])**.** *(i)* $V(x) = \sum_{j=1}^{m} \Delta p_j \, W_{K_j}(x)$. *(ii)* $W_K(x) \geq 0$ *for all* x, K. *(iii)* $V(x) \geq 0$ *for every* $n \geq 1$ *and every* $x \in [0, \infty)^n$ *if and only if* $\Delta p_j \geq 0$ *for all* j. *(iv)* $W_K(x) = 0$ *iff there are no* t, u *with* $x_t < K < x_u$.

Proof. (i) The linear term contributes $p_0 \sum_t x_t - n p_0 \bar{x} = 0$. (ii) $z \mapsto (z - K)_+$ is convex; apply Jensen's inequality to the uniform distribution on $\{x_t\}$. (iii) Sufficiency follows from (i),(ii). For necessity suppose $\Delta p_i < 0$. Choose $0 < \varepsilon < \min\{K_i,\ \min_{j \neq i} |K_j - K_i|\}$ and $x = (K_i - \varepsilon, K_i + \varepsilon)$. For $j \neq i$ both points lie on the same side of K_j, so $W_{K_j}(x) = 0$ by (iv); and $W_{K_i}(x) = \varepsilon - 0 = \varepsilon$. Hence $V(x) = \Delta p_i \varepsilon < 0$. (iv) follows from Proposition 2(b): if $\bar{x} \geq K$ then $W_K = \sum_t (K - x_t)_+$, which vanishes iff all $x_t \geq K$; if $\bar{x} < K$ then $W_K = \sum_t (x_t - K)_+$, which vanishes iff all $x_t \leq K$. In both cases this is equivalent to the absence of a strict straddle. □

Proposition S2.1 (Unequal period lengths)**.** *Let period* t *have length* $L_t > 0$ *days and define the prorated charge* $B_L(x) = \frac{L}{30} B\big(\frac{30x}{L}\big)$ *(thresholds scale by* $L/30$*). If* B *is convex then*

$$V_L(x) = \sum_t B_{L_t}(x_t) - B_{\sum_t L_t}\Big(\sum_t x_t\Big) \ \geq \ 0.$$

Proof. $g(x, L) = LB(x/L)$ is the perspective of a convex function, hence jointly convex on $L > 0$, and it is positively homogeneous of degree one [7]. For such g, $g(u+v) = 2g\big(\frac{u+v}{2}\big) \leq g(u) + g(v)$, so g is subadditive; induction gives $\sum_t g(x_t, L_t) \geq g(\sum_t x_t, \sum_t L_t)$. Scaling by $1/30$ gives the claim. With equal L_t the inequality reduces to $V \geq 0$. □

Remark 2. Using unprorated monthly thresholds with calendar months of different lengths is *not* the same comparison; in the synthetic panel it raised the mean penalty from Rs 109.3 to Rs 114.0.

S3 Exact absolute-deviation and minority-side representations

Proposition 2 (Exact identities)**.** *For every $x \in \mathbb{R}^n$ and $K \in \mathbb{R}$:*

$$(a) \quad W_K(x) = \tfrac{1}{2}\Big[\sum_t |x_t - K| - n|\bar{x} - K|\Big]; \tag{S3.1}$$

$$(b) \quad W_K(x) = \begin{cases} \sum_t (K - x_t)_+, & \bar{x} \geq K, \\ \sum_t (x_t - K)_+, & \bar{x} \leq K. \end{cases} \tag{S3.2}$$

In particular, for a two-tier tariff $V = \frac{\Delta p}{2}\big[\sum_t |x_t - K| - n|\bar{x} - K|\big]$.

Proof. (a) Substitute $(z)_+ = \frac{1}{2}(z + |z|)$ in both terms of W_K; the linear parts give $\frac{1}{2}[\sum_t (x_t - K) - n(\bar{x} - K)] = 0$. (b) If $\bar{x} \leq K$, the second term of W_K is zero. If $\bar{x} \geq K$, $W_K = \sum_t [(x_t - K)_+ - (x_t - K)] = \sum_t (K - x_t)_+$ because $(z)_+ - z = (-z)_+$. At $\bar{x} = K$ both forms agree since $\sum_t (x_t - K) = 0$. □

Lemma S3.1 (Reflection invariance)**.** *$W_K(x) = W_K(2K - x)$, where $(2K - x)_t = 2K - x_t$. Hence results proved for $\bar{x} \geq K$ transfer to $\bar{x} \leq K$.*

Proof. Each term of (S3.1) is invariant under $x_t \mapsto 2K - x_t$. □

Numerical verification. The identities were checked on 3,000 random vectors (gamma, discrete, Pareto and constant draws, $1 \leq n \leq 29$) to absolute error below 10^{-7} (`test_theory.py`), and against an independent slab-by-slab bill calculator for every synthetic household.

S4 Moment bounds

Proposition 3 (MAD, Scarf-type and SD bounds; known)**.** *For every $x \in \mathbb{R}^n$,*

$$W_K(x) \;\leq\; \frac{n}{2} \min\Big\{\mathrm{MAD},\ \sqrt{s^2 + d^2} - |d|\Big\} \;\leq\; \frac{n}{2}\, s.$$

Proof. By the triangle inequality $|x_t - K| \leq |x_t - \bar{x}| + |d|$, so (S3.1) gives $W_K \leq \frac{n}{2}\mathrm{MAD}$. By Cauchy–Schwarz, $\frac{1}{n}\sum_t |x_t - K| \leq \big(\frac{1}{n}\sum_t (x_t - K)^2\big)^{1/2} = \sqrt{s^2 + d^2}$, giving the second term. Finally $\mathrm{MAD} \leq s$ (Cauchy–Schwarz) and $\sqrt{s^2 + d^2} \leq s + |d|$. □

The second term, divided by n, is $\sup\{\mathbb{E}\,(X - K)_+ - (\mathbb{E}X - K)_+\}$ over all laws with mean $\bar{x}$ and variance s^2: Scarf's inventory bound [2], Lo's option bound [3], see also [4]. With support $[0, \infty)$ the sharp distributional bound is [3, 5]

$$\sup_{X \geq 0} \mathbb{E}\,(X - K)_+ = \begin{cases} \bar{x} - K\dfrac{\bar{x}^2}{\bar{x}^2 + s^2}, & K \leq \dfrac{\bar{x}^2 + s^2}{2\bar{x}}, \\ \frac{1}{2}\big[(\bar{x} - K) + \sqrt{s^2 + (\bar{x} - K)^2}\big], & \text{otherwise.} \end{cases}$$

Proposition S4.1 (Equality cases)**.** *(i) $W_K = \frac{n}{2}s$ iff $K = \bar{x}$, n is even and exactly half the x_t equal $\bar{x} + s$ and half equal $\bar{x} - s$ (or $s = 0$). (ii) $W_K = \frac{n}{2}\mathrm{MAD}$ iff $K = \bar{x}$ or $s = 0$ (Proposition 4). (iii) For fixed n, the Scarf-type term is attained by some $x \in \mathbb{R}^n$ with the given $(\bar{x}, s)$ iff either $s = 0$ or $np^* \in \{1, \ldots, n - 1\}$, where $p^* = \frac{1}{2}\big(1 - |d|/\sqrt{s^2 + d^2}\big)$.*

Proof. (i) Equality in $W_K \leq \frac{n}{2}\mathrm{MAD} \leq \frac{n}{2}s$ requires $\mathrm{MAD} = s$, i.e. all $|x_t - \bar{x}|$ equal to s, and $W_K = \frac{n}{2}\mathrm{MAD}$, i.e. $K = \bar{x}$ when $s > 0$ (Proposition 4). A zero-mean deviation vector with entries $\pm s$ has equally many of each sign. (ii) is Proposition 4. (iii) By Theorem 1 the maximum over n-vectors is $\max_j A_j$. Write $A_j = nf(j/n)$ with $f(p) = s\sqrt{p(1-p)} - p|d|$, which is strictly concave on $(0, 1)$ when $s > 0$ and has unique maximiser p^* with $f(p^*) = \frac{1}{2}(\sqrt{s^2 + d^2} - |d|)$; substituting $p = \frac{1-u}{2}$ gives $f'(p) = 0 \iff u/\sqrt{1 - u^2} = |d|/s$. Hence $\max_j A_j = nf(p^*)$ iff $p^* \in \{1/n, \ldots, (n-1)/n\}$. □

Example 1. $n = 3$, $\bar{x} = 280$, $s = 120$, $K = 150$: the Scarf-type bound is 70.38 kWh but $\max_j A_j = A_1 = 39.71$ kWh, attained by $x = (110.29, 364.85, 364.85)$.

S5 Finite-sample sharp bound (Theorem 1)

Theorem 1 (Proved here)**.** *Let $n \geq 2$, $\bar{x} \in \mathbb{R}$, $s \geq 0$, $K \in \mathbb{R}$ and $A_j = s\sqrt{j(n-j)} - j|\bar{x} - K|$. For every $x \in \mathbb{R}^n$ with mean $\bar{x}$ and population standard deviation s,*

$$W_K(x) \;\leq\; U^{\text{fin}} := \max\Big\{0, \max_{1 \leq j \leq n-1} A_j\Big\},$$

and the bound is attained.

Proof. By Corollary S3.1 assume $d = \bar{x} - K \geq 0$. Let $J = \{t : x_t < K\}$ and $j = |J|$. If $j = 0$ then $W_K = 0$ by (S3.2). Since $\bar{x} \geq K$, $j \leq n - 1$. By (S3.2), $D := W_K = \sum_{t \in J}(K - x_t) > 0$. Since $\sum_t (x_t - K) = nd$, the complementary group has total excess $E = \sum_{t \notin J}(x_t - K) = D + nd \geq 0$. Cauchy–Schwarz within each group gives

$$n(s^2 + d^2) = \sum_t (x_t - K)^2 \;\geq\; \frac{D^2}{j} + \frac{(D + nd)^2}{n - j} =: h(D).$$

The function h is strictly increasing on $[0, \infty)$. Multiplying $h(D) = n(s^2 + d^2)$ by $j(n - j)/n$ gives $D^2 + 2jdD + jnd^2 - j(n - j)(s^2 + d^2) = 0$, whose nonnegative root is

$$D_j = -jd + \sqrt{j^2 d^2 - jnd^2 + j(n - j)(s^2 + d^2)} = -jd + s\sqrt{j(n - j)} = A_j,$$

since $j^2 - jn + j(n - j) = 0$. Hence $W_K = D \leq A_j \leq \max_j A_j$.

Attainment. If $\max_j A_j \leq 0$, any x with the given moments has $W_K = 0$ (the argument above shows $D \leq A_j \leq 0$ is impossible for $D > 0$), and the bound 0 is attained. Otherwise let j^* maximise A_j and set $x_t = K - A_{j^*}/j^*$ for $t \leq j^*$ and $x_t = K + (A_{j^*} + nd)/(n - j^*)$ for $t > j^*$. Then $\sum_t (x_t - K) = nd$ and $\sum_t (x_t - K)^2 = h(A_{j^*}) = n(s^2 + d^2)$, so the mean is $\bar{x}$ and the standard deviation is s; the first j^* points lie strictly below K and the rest weakly above, so $W_K(x) = A_{j^*}$ by (S3.2). □

Corollary 1 (Zero-penalty certificate; Samuelson [6])**.** *If $|\bar{x} - K| \geq s\sqrt{n - 1}$ then $W_K(x) = 0$ for every x with these moments.*

Proof. $A_j/j = s\sqrt{(n - j)/j} - |d|$ is strictly decreasing in j when $s > 0$, so $\max_j A_j \leq 0$ iff $A_1 \leq 0$ iff $|d| \geq s\sqrt{n - 1}$. (Equivalently, no single observation can lie more than $s\sqrt{n - 1}$ from the mean.) □

Corollary S5.1. *$U^{\text{fin}} \leq \frac{n}{2}(\sqrt{s^2 + d^2} - |d|)$, with equality characterised in Proposition S4.1(iii). The Scarf-type bound is positive whenever $s > 0$; it therefore never certifies $W_K = 0$.*

Remark 3 (Sample versus population standard deviation)*.* If the sample standard deviation $s' = s\sqrt{n/(n - 1)}$ is substituted for s, the bounds remain valid (all are nondecreasing in s) but are no longer sharp. Mixing conventions inside a proof is an error; the code uses s throughout (`theory.pop_sd`).

S6 Threshold sensitivity (Proposition 4)

Proposition 4 (Proved here; elementary)**.** *Fix $x \in \mathbb{R}^n$ with $s > 0$, and let $m_- = \min_t x_t < \bar{x} < m_+ = \max_t x_t$. Then $K \mapsto W_K(x)$ is continuous and piecewise linear, equals 0 for $K \leq m_-$ and $K \geq m_+$, is strictly increasing on $[m_-, \bar{x}]$ and strictly decreasing on $[\bar{x}, m_+]$. Hence $K = \bar{x}$ is the unique maximiser and $\max_K W_K(x) = \frac{n}{2}\text{MAD}$.*

Proof. For $K < \bar{x}$, by (S3.2) $W_K = \sum_t (K - x_t)_+$, whose right and left derivatives are $\#\{t : x_t \le K\}$ and $\#\{t : x_t < K\}$; both are ≥ 1 on $(m_-, \bar{x})$ and both are 0 for $K < m_-$. For $K > \bar{x}$, $W_K = \sum_t (x_t - K)_+$, with derivatives $-\#\{t : x_t > K\}$ and $-\#\{t : x_t \ge K\}$, both ≤ -1 on $(\bar{x}, m_+)$. Continuity at $\bar{x}$ follows from (S3.1). Finally $W_{\bar{x}} = \frac{1}{2}\sum_t |x_t - \bar{x}| = \frac{n}{2}\mathrm{MAD}$. □

Remark 4 (Plateaus, kinks and ties). W_K has kinks at every distinct x_t and at $\bar{x}$; it is flat (and zero) only outside $[m_-, m_+]$. There are no ties for the maximiser when $s > 0$. If $s = 0$, $W_K \equiv 0$ and every K is a maximiser. The result fixes Δp; in a tariff-design problem moving K changes the revenue-neutral prices, so the maximiser of the *monetary* penalty over designs need not be $\bar{x}$.

S7 Non-negative consumption (Theorem 2)

Theorem 2 (Validity proved here)**.** *If in addition $x_t \ge 0$ for all t, then*

$$W_K(x) \le \max\Big\{0, \max_{1\le j\le n-1} \min\{A_j, c_j\}\Big\}, \qquad c_j = \begin{cases} jK, & \bar{x} \ge K, \\ n\bar{x} - jK, & \bar{x} < K. \end{cases}$$

Proof. Case $\bar{x} \ge K$: with J as in Theorem 1, $W_K = \sum_{t\in J}(K - x_t) \le jK$ because $x_t \ge 0$, and $W_K \le A_j$ by Theorem 1. Case $\bar{x} < K$: let $J^+ = \{t : x_t > K\}$ with $|J^+| = j \le n - 1$. Then $W_K = \sum_{t\in J^+}(x_t - K) = \sum_{t\in J^+} x_t - jK \le \sum_t x_t - jK = n\bar{x} - jK$, using $x_t \ge 0$ for $t \notin J^+$. Applying Theorem 1 to the reflected vector gives $W_K \le A_j$. □

Conjecture 1 (Sharpness)**.** *The bound of Theorem 2 is attained by some nonnegative x with the given moments whenever such vectors exist (i.e. $s \le \bar{x}\sqrt{n-1}$).*

Evidence. Multistart SLSQP maximisation of W_K under the moment and nonnegativity constraints matched the bound in all 28 structured cases ($n \in \{3, 4, 6, 12\}$, seven $(\bar{x}, s, K)$ settings) and in 120 random cases ($2 \le n \le 6$), with maximum relative gap 8×10^{-14}. A proof would have to show that, when c_j binds, the remaining variance can always be absorbed by the majority group; we have not completed this argument.

S8 Multi-tier tariffs: decomposition and the joint bound (Theorem 3)

Theorem 3 (Proved here)**.** *Let B be convex with prices $p_0 \le \cdots \le p_m$ and intercepts c_k as in* (S1.2)*. For a count vector $N = (n_0, \ldots, n_m) \in \mathbb{Z}^{m+1}_{\ge 0}$ with $\sum_k n_k = n$, let $\bar{p}_N = \frac{1}{n}\sum_k n_k p_k$ and $\sigma_N^2 = \frac{1}{n}\sum_k n_k (p_k - \bar{p}_N)^2$. Then for every $x \in \mathbb{R}^n$ with mean $\bar{x}$ and population standard deviation s,*

$$V(x) \;\le\; U^{\mathrm{joint}} := \max_N \Big[n\bar{x}\,\bar{p}_N + ns\,\sigma_N - \sum_k n_k c_k\Big] - nB(\bar{x}),$$

and the bound is attained (over $\mathbb{R}^n$). The maximum is over $\binom{n+m}{m}$ count vectors.

Proof. Let $S = \{x \in \mathbb{R}^n : \sum_t x_t = n\bar{x},\ \sum_t (x_t - \bar{x})^2 = ns^2\}$. Extending (S1.2) to all real x by $B(x) := \max_k (p_k x - c_k)$ (which agrees with the tariff on $x \ge 0$), for any assignment $\kappa : \{1, \ldots, n\} \to \{0, \ldots, m\}$,

$$\sum_t B(x_t) = \max_\kappa \sum_t \big(p_{\kappa(t)} x_t - c_{\kappa(t)}\big).$$

Hence $\sup_{x\in S} \sum_t B(x_t) = \max_\kappa \sup_{x\in S} \sum_t (a_t x_t - c_{\kappa(t)})$ with $a_t = p_{\kappa(t)}$. Writing $x_t = \bar{x} + y_t$ with $\sum_t y_t = 0$, $\sum_t y_t^2 = ns^2$, we get $\sum_t a_t x_t = n\bar{x}\bar{a} + \sum_t (a_t - \bar{a}) y_t \le n\bar{x}\bar{a} + \|a - \bar{a}\|\sqrt{n}\,s$ by

Cauchy–Schwarz, with equality at $y = \sqrt{n}\, s\,(a - \bar{a})/\|a - \bar{a}\|$ (or any feasible y if a is constant). Since $\|a - \bar{a}\| = \sqrt{n}\,\sigma_N$ and $\bar{a} = \bar{p}_N$ depend on κ only through the counts N, the supremum equals $\max_N[n\bar{x}\bar{p}_N + ns\sigma_N - \sum_k n_k c_k]$. Subtracting $nB(\bar{x})$ gives the bound; the maximising y and assignment attain it, because $\sum_t B(x_t)$ is at least the value of that assignment and at most the overall supremum. □

Corollary S8.1. *For $m = 1$, $U^{\text{joint}} = \Delta p \cdot U^{\text{fin}}$ (Theorem 1); this was also verified numerically on 2,000 random instances (maximum difference 1.2×10^{-11}).*

Remark 5 (Nonnegativity and joint versus componentwise sharpness). U^{joint} is a valid bound for nonnegative consumption; it is sharp under nonnegativity whenever the maximising vector is nonnegative, as in all examples below. By Proposition 1(i), $V \le \sum_j \Delta p_j U^{\text{fin}}(K_j)$, but the per-threshold maximisers are different vectors, so the sum is in general *not* attainable: for the reported Uttar Pradesh schedule ($K = 150, 300, 500$; $\Delta p_j = 0.5$) with $n = 12$, $\bar{x} = 280$, $s = 160$, the sum is Rs 806.39 whereas U^{joint} = Rs 565.10, attained by four readings of 53.73 kWh and eight of 393.14 kWh. Multistart optimisation (200 starts) found the same value independently.

Componentwise bounds. Summing Proposition 3 over thresholds gives $V \le \frac{n}{2}\sum_j \Delta p_j \min\{\text{MAD}, \sqrt{s^2 + d_j^2} - |d_j|\}$ and $V \le \frac{n}{2}(p_m - p_0)\, s$. The MAD version is attained iff every threshold with $\Delta p_j > 0$ equals $\bar{x}$, which is impossible for $m \ge 2$ distinct thresholds unless $s = 0$.

S9 Constrained tariff design

Consider two-tier tariffs $\theta = (p_1, \delta, K)$ with $p_2 = p_1 + \delta$ and households $h = 1, \ldots, H$ with consumption vectors $x_h \in [0, \infty)^n$. Let $W_h(K) = W_K(x_h)$, $T = \sum_{h,t} x_{ht}$, $G(K) = \sum_{h,t} (x_{ht} - K)_+$ and $\text{cov}(K)$ the share of household-periods above K.

Proposition S9.1. *(i) For fixed K, revenue $R = p_1 T + \delta G(K)$, the bill at basic consumption x_b, $p_1 \min(x_b, K) + \delta\,(x_b - K)_+$, and every $V_h = \delta W_h(K)$ are linear in (p_1, δ). Hence for fixed K the problems* $\min\ \delta \cdot \phi(K)$ *with $\phi \in \{\frac{1}{H}\sum_h W_h,\ \max_h W_h,\ \max_h U_h\}$ subject to the constraints of Section 7 of the manuscript are linear programmes in two variables. (ii) If the feasible set is nonempty, minimisers exist. (iii) For fixed K with $\phi(K) > 0$, every optimal solution has the smallest feasible δ; consequently along the frontier $\delta_{\min} \mapsto \min V_{\max}$ at fixed K the penalty is linear in δ. (iv) Without the progressivity and coverage constraints the infimum is 0, attained by $\delta = 0$, or by $K \le \min_{h,t} x_{ht}$, or by $K \ge \max_{h,t} x_{ht}$ (Proposition 1(iv)).*

Proof. (i) is immediate from (S1.1). (ii) For each fixed K the feasible set in (p_1, δ) is a compact polygon (bounded prices, linear constraints), so the LP has a minimiser whenever it is feasible. Over a finite grid of K values a global minimiser therefore exists. For continuous K, the coverage share is a right-continuous step function of K, so the feasible set need not be closed; we do not claim existence over continuous K and report grid-global optima only. (iii) The objective is $\phi(K)\delta$ with $\phi(K) > 0$. (iv) follows from $W_h(K) = 0$ in those cases. □

The computational procedure solves the LP exactly (HiGHS) at each grid value of K (step 1 kWh), so the reported optima are global *on the grid*; continuous optimality in K is not certified.

S10 Extremal configurations and edge cases

- $n = 1$: $V = 0$; the bounds are defined as 0.
- Constant consumption ($s = 0$): $V = 0$; all bounds equal 0.
- All readings on a threshold: $V = 0$ (Proposition 1(iv)).

- Readings $K \pm \varepsilon$ in equal numbers: $V = \frac{n}{2}\Delta p\,\varepsilon$, which attains the SD, MAD, Scarf and finite-sample bounds simultaneously.
- Zero readings: genuine zero consumption is admissible and is retained; zeros arising from outages must be treated as missing (data protocol), otherwise s and V are inflated.
- Extremal two-point vectors (Theorem 1) may require negative consumption when $A_{j^*}/j^* > K$ and $\bar{x} \geq K$; Theorem 2 then gives the relevant bound.

S11 Counterexamples to invalid generalisations

1. **All-units slabs.** $B^{\mathrm{AU}}(x) = 5.5x$ for $x \leq 150$, $6.0x$ for $x > 150$; $x = (140, 166)$: $\sum_t B^{\mathrm{AU}}(x_t) = 770 + 996 = 1766 < 2 \times 6.0 \times 153 = 1836$, so $V = -70$.
2. **Decreasing blocks.** $p = (6, 5)$, $K = 150$, $x = (100, 200)$: $V = -50 < 0$.
3. **Summing sharp bounds.** Section S8: Rs 806.39 versus the attainable Rs 565.10.
4. **Scarf sharp for fixed n.** Section S4 example: 70.38 versus 39.71.
5. **Longer billing periods always reduce the penalty.** Merging *nested* periods cannot increase V: if bimonthly periods are unions of monthly periods, Jensen's inequality applied within each pair gives $V_{\text{bimonthly}} \leq V_{\text{monthly}}$ (with thresholds scaled as $B_2(E) = 2B(E/2)$). Non-nested schemes are not ordered: in the synthetic panel 70 of 300 households had $V_{\text{quarterly}} > V_{\text{bimonthly}}$ (e.g. readings 186.9, 156.0, 242.8, 394.7, 435.3, 514.5, 513.6, 396.7, 404.7, 252.8, 175.5, 2 Rs 275.53 quarterly versus Rs 223.32 bimonthly), while $V_{\text{quarterly}} \leq V_{\text{monthly}}$ held for all, as nesting implies.
6. **The penalty maximiser over designs is $\bar{x}$.** False once prices adjust to keep revenue fixed (Remark in Section S6).
7. **Variance alone determines the penalty.** Two vectors with identical $(\bar{x}, s)$ can have $V = 0$ and $V > 0$: e.g. $K = 150$: $x = (160, 160, 160, 280)$ and $x' = (100, 220, 220, 220)$ both have $\bar{x} = 190$ and $s = \sqrt{2700} \approx 51.96$, but $W_K(x) = 0$ while $W_K(x') = 50$ (and x' attains Theorem 1).

S12 Numerical validation

The archive's `04_Tests` directory contains 26 executed `pytest` tests (all passing; log in `05_Computational_Resul` hand-computed bills (420 kWh ⇒ Rs 2,505), equality between the hinge and slab-by-slab implementations on 500 random values, identities and bound orderings on 3,000 random vectors, attainment of Theorems 1 and 3, the Samuelson certificate, threshold maximiser, decomposition, non-convex counterexample, unequal-period non-negativity, data-pipeline rules, optimisation feasibility and determinism. `validate.py` independently recomputes every household penalty and every reported design with the slab-by-slab calculator. Adversarial searches are described in Sections S7 and S8. Numerical checks support but do not replace the proofs.

S13 Additional derivations and methodological details

Billing-period scaling. For k merged cycles the consistent schedule is $B_k(E) = kB(E/k)$ (thresholds and fixed charges scale with k). Using the monthly threshold on a quarterly total would move almost all consumption into the top block and is not a meaningful comparison.

Observed-data protocol (not yet executed). Interval versus cumulative energy is determined from documentation and checked by monotonicity within meter; cumulative registers are

differenced within day, negative differences (resets) set missing; days with $< 90\%$ interval coverage are missing (never zero-filled); daily energy above 100 kWh is flagged implausible; months with $\geq 90\%$ valid days are retained and scaled to the calendar month; households need ≥ 10 retained months in the analysis window; penalties use prorated thresholds (Proposition S2.1); inference uses a household-cluster bootstrap stratified by district [8].

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